

Deployment of Application using Azure Virtual Machine Scale Set (VMSS)

Introduction

Cloud computing is a modern technology that allows users to access computing resources such as servers, storage, networking, and applications over the internet instead of maintaining physical hardware. It provides flexibility, scalability, security, and cost efficiency. Organizations use cloud platforms to deploy applications quickly and manage them efficiently without investing in infrastructure.

Microsoft Azure is a leading cloud computing platform developed by Microsoft. It offers various services such as Virtual Machines, databases, networking solutions, storage systems, and application hosting services. Azure allows developers and organizations to build, deploy, and manage applications in a highly available and secure environment.

A Virtual Machine (VM) is a software-based computer that runs an operating system and applications just like a physical computer. However, when an application receives high traffic, a single virtual machine may not be enough to handle the load efficiently.

To solve this problem, Azure provides a service called Virtual Machine Scale Set (VMSS). A Virtual Machine Scale Set allows users to create and manage a group of identical virtual machines. It supports automatic scaling, meaning the number of virtual machines can increase or decrease based on user demand. This ensures high availability, load distribution, and better performance.

This hands-on exercise demonstrates how to create a VM Scale Set, configure networking, deploy a custom application using a bootstrap script, configure firewall rules, and safely terminate resources.

Objective of the Hands-On

The main objectives of this exercise are:

- To create a Resource Group in Microsoft Azure.
- To understand the process of creating a Virtual Machine Scale Set.
- To deploy a custom Python application using a bootstrap script.
- To configure networking and load balancing.
- To create firewall rules to allow HTTP (Port 80) and SSH (Port 22).
- To access the deployed application through a web browser.
- To terminate cloud resources properly to avoid unnecessary billing.

Creation of Resource Group

A Resource Group in Azure acts as a logical container that holds related resources for an application. It helps in organizing and managing Azure services efficiently. All the components such as Virtual Machines, Load Balancer, Virtual Network, and storage are stored inside the resource group.

In this exercise, a resource group named ****GL-VMSS-RG**** is created. This resource group will contain all the resources related to the Virtual Machine Scale Set deployment.

Creating a resource group ensures better management, easy deletion of resources, and improved organization of cloud services.

Creation of Virtual Machine Scale Set (VMSS)

A Virtual Machine Scale Set allows deployment of multiple identical virtual machines. It ensures high availability and supports automatic scaling.

The following configuration is used:

- Region: East US
- Operating System Image: Ubuntu 20.04
- Authentication Type: SSH Public Key
- Username: azureuser
- Initial Instance Count: 2
- Scaling Policy: Custom
- Minimum Instances: 2
- Maximum Instances: 5

The initial instance count is set to 2, meaning two virtual machines are created initially. The scaling policy is set to custom, which allows scaling between a minimum of 2 instances and a maximum of 5 instances depending on workload.

This configuration ensures that the application can handle increased traffic efficiently.

Networking Configuration

Networking plays an important role in cloud deployments. In this step:

- A new Virtual Network is created.
- A Load Balancer is enabled.
- A Backend Pool is configured.
- Network Interface is selected.

The Load Balancer distributes incoming user traffic across multiple virtual machines. This prevents overloading of a single machine and improves availability and performance.

The Virtual Network provides secure communication between Azure resources. The architecture is:

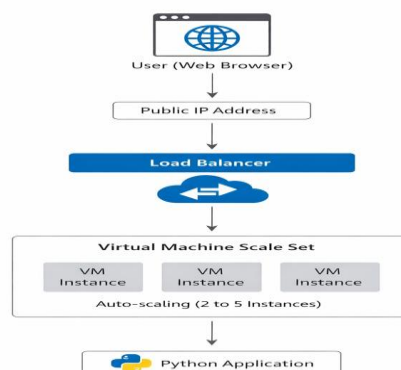


Figure 1: Architecture of Azure VM Scale Set Deployment

The above architecture shows how user requests are routed through the load balancer and distributed across multiple VM instances in the scale set.

Scaling Configuration

Scaling ensures that the application adjusts automatically based on demand. In this configuration:

- Initial Instances: 2
- Minimum Instances: 2
- Maximum Instances: 5

If user traffic increases, the system can increase the number of virtual machines up to 5. If traffic decreases, the number of instances can reduce but will not go below 2.

This improves cost efficiency and ensures performance stability.

Health Monitoring Configuration

Application Health Monitoring is enabled to ensure that the application is running correctly on all virtual machines.

Configuration includes:

- Protocol: HTTP
- Port: 80
- Path: /health.html

The health check monitors whether the application is responding properly. If a virtual machine becomes unhealthy, Azure can replace it automatically. This ensures reliability and continuous availability.

Deployment using Bootstrap Script

A bootstrap script is added under the Custom Data section while creating the VM Scale Set. This script runs automatically when the virtual machine starts.

The script performs the following tasks:

- Updates the Ubuntu system.
- Installs Python and required packages.
- Downloads the application ZIP file.
- Extracts the application files.
- Installs required dependencies.
- Starts the Python application in the background.

This automation removes the need for manual installation and configuration. It ensures that the application is deployed automatically on all VM instances.

Firewall Configuration (Inbound Rules)

To allow external users to access the application, inbound port rules are configured.

The following ports are opened:

- Port 80 (HTTP) – Allows web traffic.
- Port 22 (SSH) – Allows secure remote login to the virtual machine.

Port 80 allows users to access the application through a web browser.

Port 22 allows administrators to connect securely using SSH for maintenance purposes.

Proper firewall configuration ensures security while allowing required access.

Accessing the Application

After successful deployment:

- Navigate to the Load Balancer.
- Copy the Public IP address.
- Paste the IP address into a web browser.

If configuration is correct, the custom application loads successfully. The Load Balancer ensures that traffic is distributed between multiple virtual machines.

Termination of Resources

After completing the exercise, it is important to delete the resources to avoid unnecessary billing.

Steps include:

- Navigate to Resource Groups.
- Select GL-VMSS-RG.
- Click Delete Resource Group.
- Confirm deletion.

Deleting the resource group removes all associated services such as VMSS, Load Balancer, and Virtual Network.

This ensures cost control and proper cloud resource management.

Conclusion

This hands-on exercise demonstrates the deployment of a scalable and highly available application using Azure Virtual Machine Scale Set. It covers essential cloud concepts such as resource grouping, scaling, load balancing, health monitoring, automation using bootstrap scripts, and security configuration.

Virtual Machine Scale Sets are useful for applications that require high performance, automatic scaling, and reliability. Azure simplifies infrastructure management and allows developers to focus on application development rather than hardware maintenance.

This exercise provides practical understanding of how modern cloud-based applications are deployed and managed efficiently.